



Graphs in Machine Learning

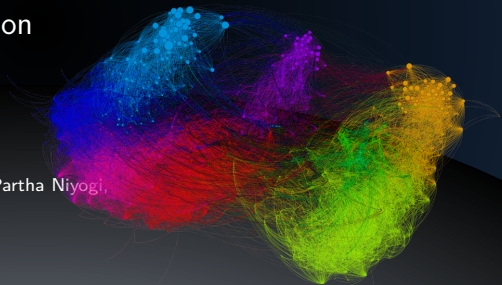
SSL with Graphs: Harmonic Functions

Gaussian Random Fields Solution

Michal Valko

Isara Labs

Partially based on material by: Mikhail Belkin, Partha Niyogi,
Olivier Chapelle, Bernhard Schölkopf



SSL with Graphs: Harmonic Functions

Zhu/Ghahramani/Lafferty: Semi-Supervised Learning Using Gaussian Fields and Harmonic Functions (ICML 2013)

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The harmonic solution is obtained from the mincut one ...

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$$\min_{\mathbf{f} \in \{\pm 1\}^{n_l+n_u}} \propto \sum_{i=1}^{n_l} (f(\mathbf{x}_i) - y_i)^2 + \lambda \sum_{i,j=1}^{n_l+n_u} w_{ij} (f(\mathbf{x}_i) - f(\mathbf{x}_j))^2$$

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$$\min_{\mathbf{f} \in \mathbb{R}^{n_l + n_u}} \sum_{i,j=1}^{n_l + n_u} w_{ij} (f(\mathbf{x}_i) - f(\mathbf{x}_j))^2$$

$$\text{s.t. } y_i = f(\mathbf{x}_i) \quad \forall i = 1, \dots, n_l$$

SSL with Graphs: Harmonic Functions

Properties of the relaxation from ± 1 to $\mathbb{R}$

- there is a closed form solution for $\mathbf{f}$
- this solution is unique
- globally optimal

SSL with Graphs: Harmonic Functions

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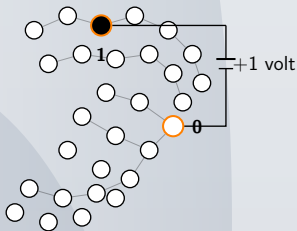
SSL with Graphs: Harmonic Functions

Properties of the relaxation from ± 1 to $\mathbb{R}$

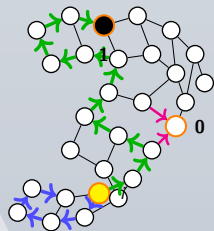
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- electric-network interpretation
- random-walk interpretation

SSL with Graphs: Harmonic Functions

$$R_{ij} = \frac{1}{w_{ij}}$$



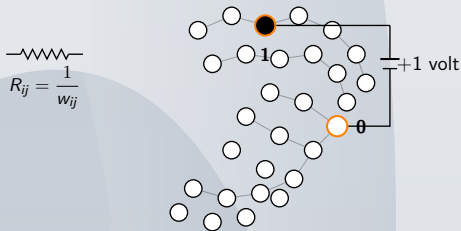
(a) The electric network interpretation



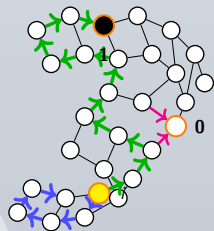
(b) The random walk interpretation

Figure from zhu2003semi-supervised<empty citation>

SSL with Graphs: Harmonic Functions



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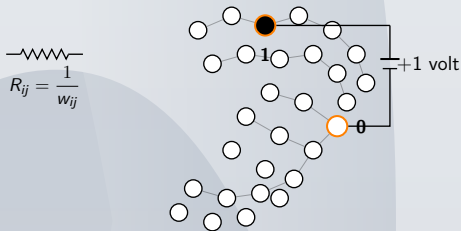
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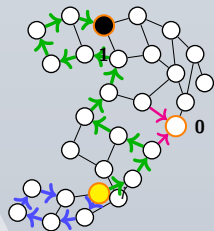
Random walk interpretation:

1) start from the vertex you want to label and randomly walk

SSL with Graphs: Harmonic Functions



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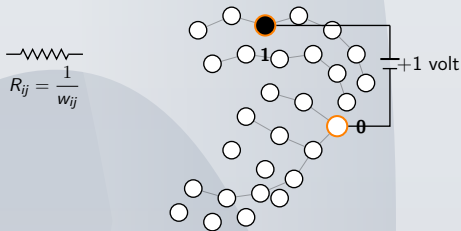
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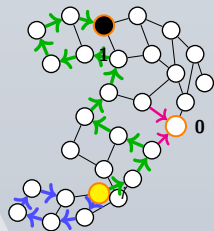
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$$2) P(j|i) = \frac{w_{ij}}{\sum_k w_{ik}}$$

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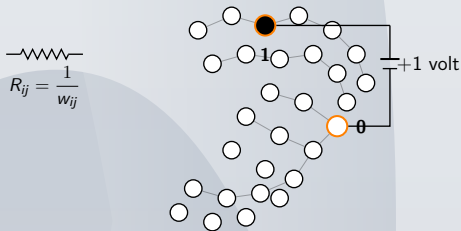
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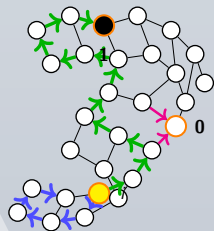
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SSL with Graphs: Harmonic Functions



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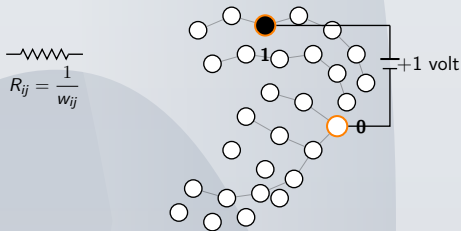
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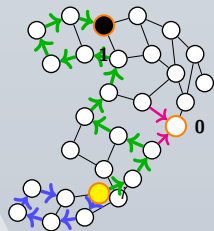
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- 2) $P(j|i) = \frac{w_{ij}}{\sum_k w_{ik}} \equiv \mathbf{P} = \mathbf{D}^{-1}\mathbf{W}$
- 3) finish when a labeled vertex is hit

SSL with Graphs: Harmonic Functions



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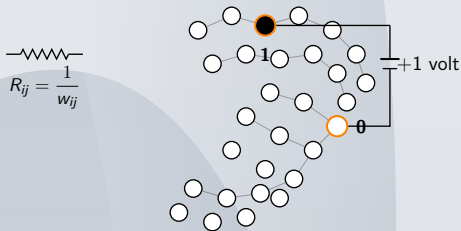
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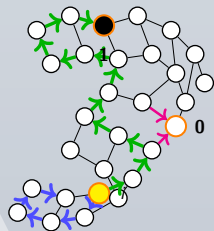
absorbing random walk

$$f_i =$$

SSL with Graphs: Harmonic Functions



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absorbing random walk

f_i = probability of reaching a positive labeled vertex

SSL with Graphs: Harmonic Functions

How to compute HS?

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How to compute HS? **Option A:** iteration

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Properties:

- this will converge to the harmonic solution

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Properties:

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- we can set the initial values for unlabeled nodes arbitrarily
- an interesting option for large-scale data

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How to compute HS? **Option B:** Closed form solution

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$$\Omega(\mathbf{f}) =$$

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$\mathbf{L}$ is a $(n_l + n_u) \times (n_l + n_u)$ matrix:

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SSL with Graphs: Harmonic Functions

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Yes, Lagrangian multipliers are an option, but . . .

SSL with Graphs: Harmonic Functions

Let us compute **harmonic** solution using **harmonic** property!

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SSL with Graphs: Harmonic Functions

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In matrix notation

$$\begin{bmatrix} \mathbf{L}_{ll} & \mathbf{L}_{lu} \\ \mathbf{L}_{ul} & \mathbf{L}_{uu} \end{bmatrix} \begin{bmatrix} \mathbf{f}_l \\ \mathbf{f}_u \end{bmatrix} = \begin{bmatrix} \cdots \\ \mathbf{0}_u \end{bmatrix}$$

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...from which we get

$$\mathbf{f}_u = \mathbf{L}_{uu}^{-1}(-\mathbf{L}_{ul}\mathbf{f}_l)$$

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Note that $\mathbf{f}_u$ does not depend on $\mathbf{L}_{ll}$.

SSL with Graphs: Harmonic Functions

Can we see that this calculates the probability of a random walk?

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Split the equation into +ve & -ve part:

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Michal Valko

`michal.valko@inria.fr`

Isara Labs



<https://misovalko.github.io/mva-ml-graphs.html>